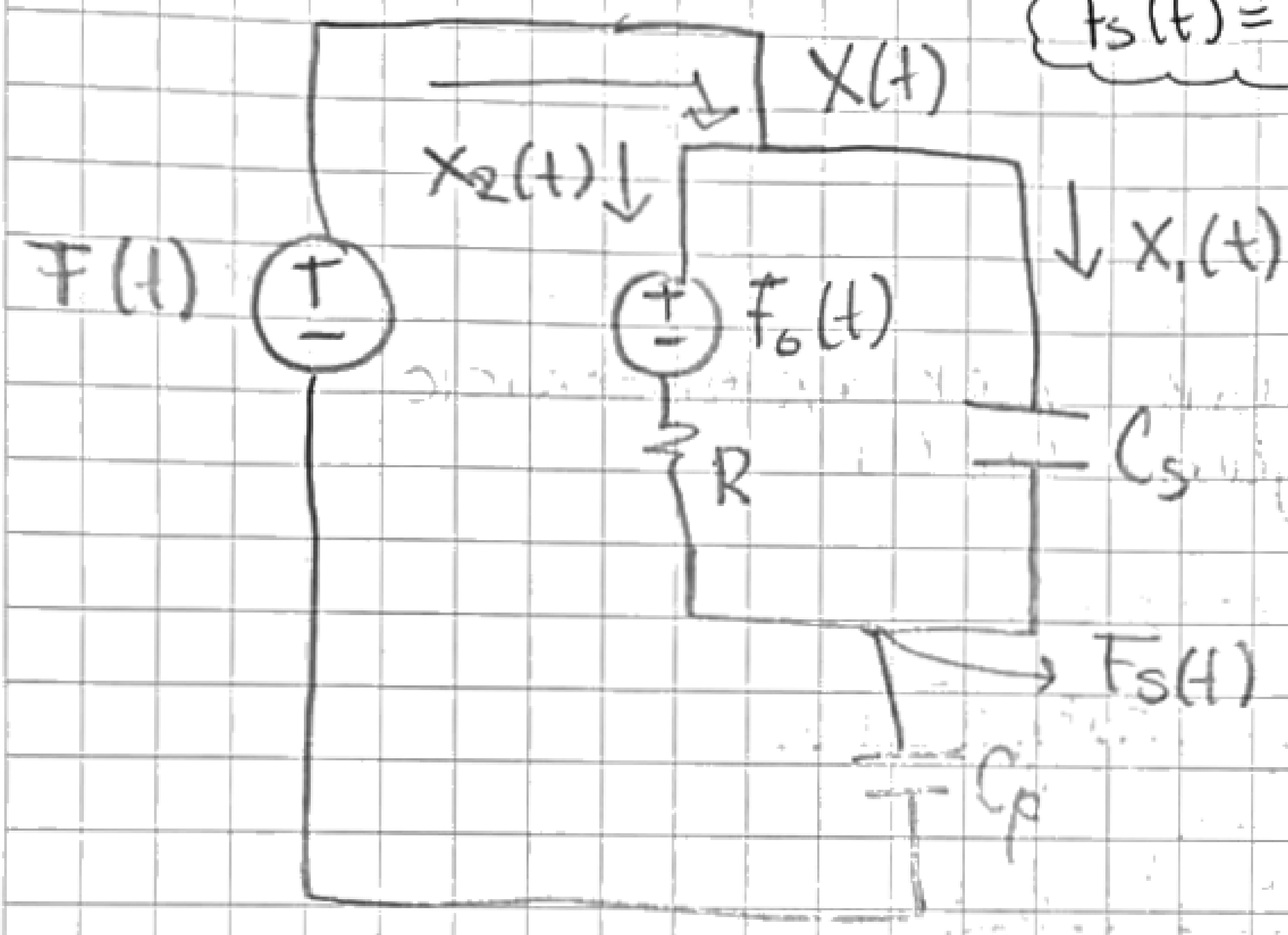
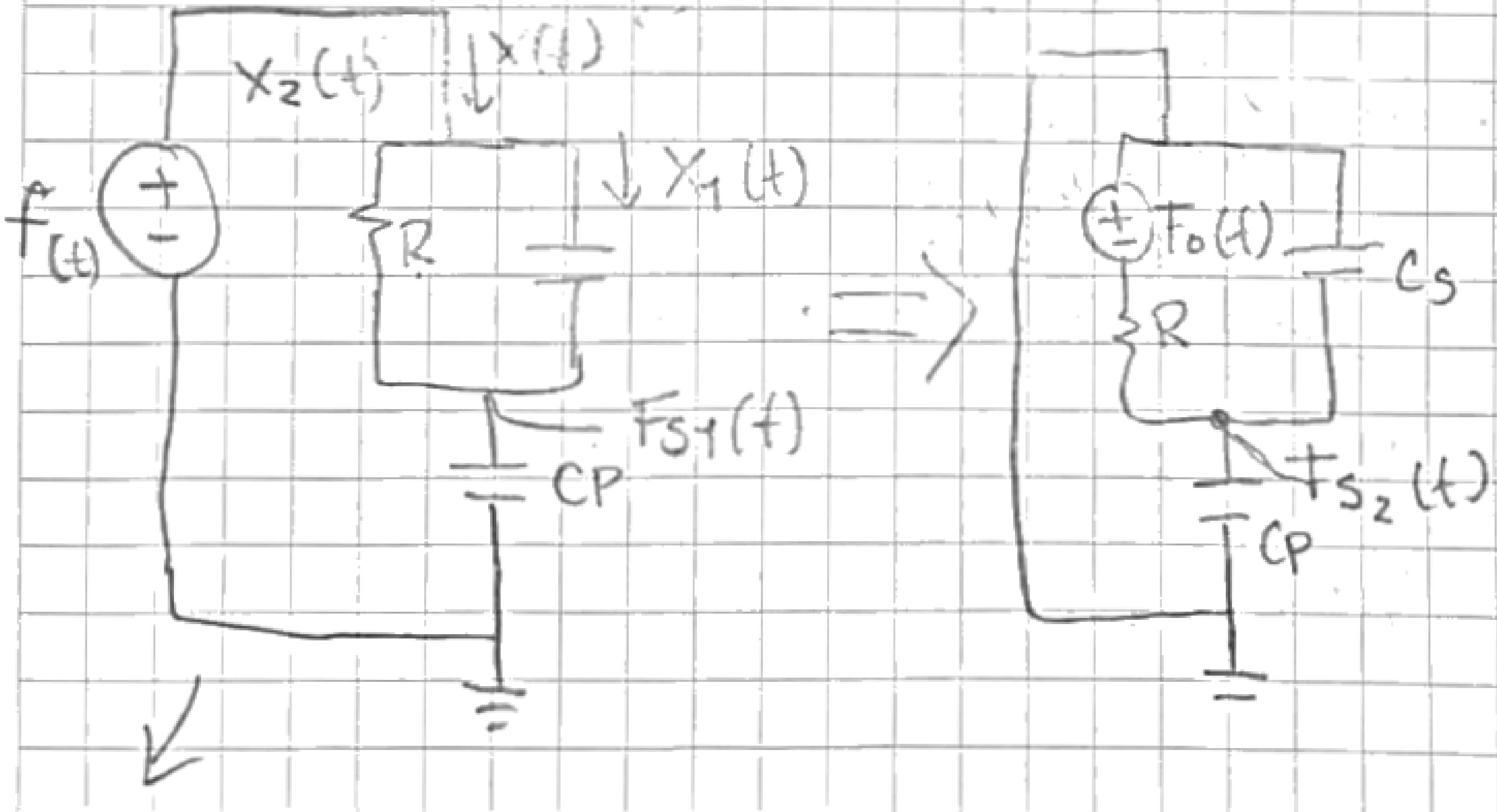
Miercoles 28 de Octubre del 2026

Practica 3

Modelado y control del sistema musculoesquelético

$$F_s(t) = F_{s1}(t) + F_{s2}(t)$$





Circuito 1

FIRST CLAS

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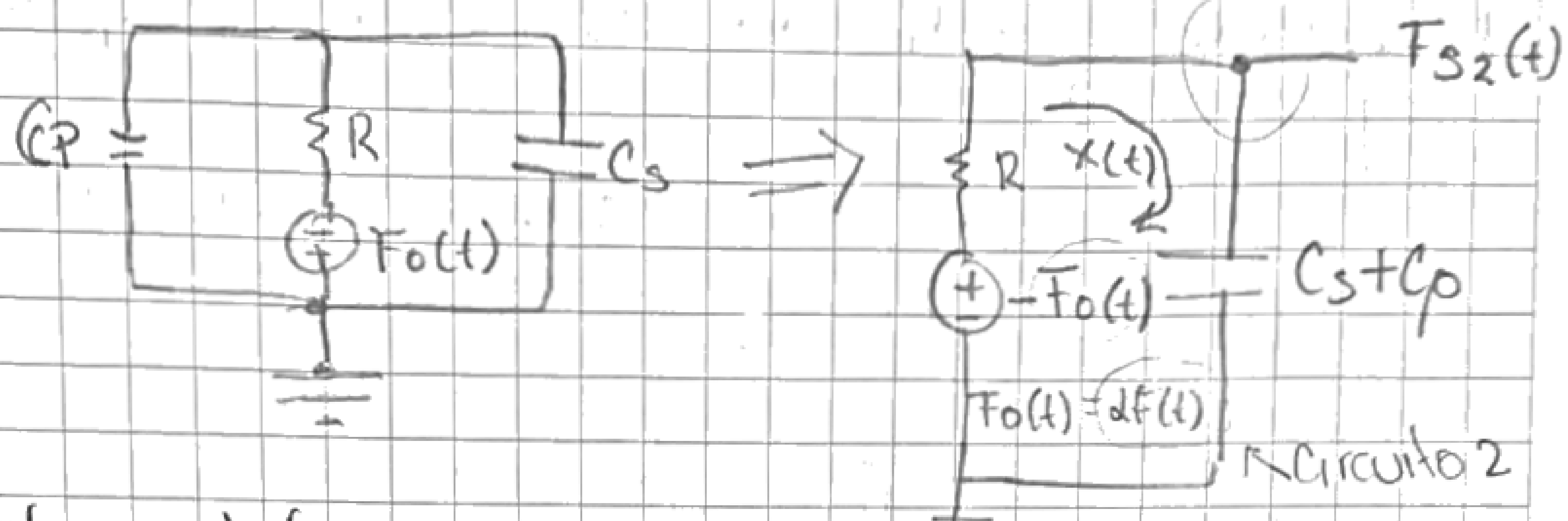

- Análisis del circuito por superposición

∘ Circuito 1°: Apagando Fo(t)

I = lujos

$x(t) = x_1(t) + x_2(t)$

$x_1(t) =$

$x_1(t) = C_s \frac{d[F(t) - F_{s1}(t)}{dt}$

$x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$

$\to C_p \frac{dF_{s1}(t)}{dt}$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s S[F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) = [C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$(C_p S + C_s S + \frac{1}{R}) F_{s1}(s) = (C_s S + \frac{1}{R}) F(s)$$

$$\frac{R(C_p S + C_s)S + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

Salida 1

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R} F(s)}{\frac{R(C_p + C_s)s + 1}{R}} = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

First Class

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Circuito 2: Apagando $F(t)$

$$-\alpha F(t) = Rx(t) + \frac{1}{C_p+C_s}\int x(t)dt$$

$$F_{s2}(t) = \frac{1}{C_p+C_s}\int x(t)dt$$

$$-\alpha F(s) = RX(s) + \frac{X(s)}{(C_p+C_s)s}$$

$$F_{s2}(S) = \frac{X(s)}{(C_p+C_s)S}$$

$$F(s) = -\frac{X(s)}{\alpha}\,\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}$$

$$\frac{F_{s2}(s)}{F(s)} = \frac{\frac{X(s)}{(C_p+C_s)s}}{-\frac{X(s)}{\alpha}\,\frac{R(C_p+C_s)s+1}{(C_p+C_s)s}} = -\frac{\alpha}{R(C_p+C_s)s+1}$$

$$F_{s2}(S) = \frac{\alpha F(s)}{R(C_p+C_s)s+1}$$

Funcion de transferencia

$$F_s(s) + F_{s1}(s) + F_{s2}(s) = \frac{C_sR_s+1}{R(C_p+C_s)s+1}F(s) - \frac{\alpha F(s)}{R(C_p+C_s)s+1}$$

$$F_s(s) = \frac{C_sR_s+1-\alpha}{R(C_p+C_s)s+1}F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_sR_s+1-\alpha}{R(C_p+C_s)s+1}$$

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